A Digital Twin-Based State Estimation Framework for Decision Support in Mining Operations

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***Abstract*—Modern mining operations require decision-support systems that combine real-time data, physical process understanding, and predictive simulation. A major limitation of many existing approaches is their dependence on incomplete or noisy observations, which can lead to unrealistic system representations and poor downstream decision-making. This paper proposes a digital twin-based framework for modeling and estimating the state of mining operations through the integration of dynamic system modeling, sensor/SCADA data, and state estimation techniques such as Kalman filtering. The resulting estimated state can be consumed by simulation and AI-based decision layers to support more reliable operational recommendations. The paper is intentionally structured as an initial collaborative draft: the student is expected to further develop the modeling, implementation, and experimental sections; the supervising professor is expected to comment on novelty, scientific rigor, and publication strategy; and the senior co-author provides the cyber-physical systems framing and industrial motivation.**

***Index Terms*—Digital Twin, State Estimation, Kalman Filter, Mining Operations, Decision Support, Cyber-Physical Systems**

I. Introduction

The mining sector is undergoing a rapid digital transformation, driven by the need to improve productivity, safety, energy efficiency, and operational resilience. In this context, digital twins, industrial simulation, and artificial intelligence have emerged as promising tools for supporting operational decision-making.

However, many AI- or simulation-driven systems assume that the operational state of the process is fully known. In practice, mining systems are only partially observable: measurements are noisy, delayed, or incomplete, and several variables of interest are not directly sensed. This creates a gap between physical reality and digital decision layers.

This paper addresses that gap by proposing a framework centered on **state estimation for a digital twin of mining operations**. The central idea is that a decision-support architecture becomes significantly more reliable when its simulation and AI modules rely on an estimated, physically grounded state rather than on raw observations alone.

The contribution of this work is threefold:

1) A dynamic-systems formulation of a mining operation suitable for digital twin development.
2) A state estimation layer that fuses sensor/SCADA information with a process model.
3) A modular architecture that allows the estimated state to feed downstream simulation and AI-based decision engines.

Comment for Omar: strengthen the novelty claim relative to mining digital twins and cyber-physical decision systems literature.

Task for Daniel: contextualize the exact mining sub-process to be studied (e.g., transport, comminution support process, dispatch, or integrated material flow).

II. Related Work

Digital twins have become increasingly relevant in industrial domains, including manufacturing, energy, logistics, and mining. In mining, they have been used for equipment monitoring, process optimization, predictive maintenance, and scenario analysis. At the same time, state estimation methods such as the Kalman Filter and its nonlinear variants have long been used in control, signal processing, and robotics to recover latent system variables from noisy measurements.

Despite these developments, there remains a practical disconnect between:

- physical process models,
- industrial sensing infrastructure,
- state estimation,
- simulation, and
- AI-based decision support.

Many proposed decision systems use data-driven prediction or optimization without explicitly enforcing a physically interpretable state representation. For large-scale industrial operations, especially mining, this can limit robustness and trustworthiness.

This work contributes by framing state estimation not as an isolated control problem, but as the enabling layer for a digital twin-driven decision-support system.

Task for Daniel: add 10–15 references on mining digital twins, SCADA-based industrial modeling, and Kalman filtering in industrial systems.

Comment for Omar: refine the literature gap so the paper is positioned either for an industrial AI venue or a systems/control venue.

III. Problem Formulation

Consider a mining operation represented as a partially observable dynamic system. Let x_k denote the internal state of the process at time step k, u_k the operational inputs or decisions, and y_k the measured outputs from sensors and SCADA infrastructure.

The system can be described as:

$$x_{k+1} = f(x_k, u_k) + w_k \tag{1}$$

$$y_k = h(x_k) + v_k \tag{2}$$

where w_k and v_k denote process and measurement noise, respectively.

Examples of state variables may include:

- material accumulation levels,
- equipment load conditions,
- flow rates or effective throughput states,
- queue or congestion indicators,
- latent operational delays.

Examples of inputs may include:

- equipment dispatching decisions,
- routing choices,
- production setpoints,
- scheduling changes.

The main challenge is that not all variables in x_k are directly observed. Therefore, a state estimator must reconstruct the most consistent estimate $\hat{x}_k$ from the available measurements.

Task for Daniel: instantiate this formulation with a concrete use case and variable definitions.

IV. Proposed Framework

The proposed framework consists of four main layers:

A. Physical Process Model

A simplified but interpretable dynamic model of the mining process is developed. This model should be expressive enough to capture relevant physical relationships while remaining tractable for estimation and simulation.

B. Data Acquisition and Preprocessing

Sensor and SCADA data are collected, synchronized, filtered, and mapped to the variables used by the model. This stage may include missing data handling, resampling, outlier filtering, and unit standardization.

C. State Estimation Layer

A Kalman Filter or related estimator is used to combine the process model with sensor observations. In a linear setting, the estimator can be expressed as:

$$\hat{x}_{k|k-1} = A\hat{x}_{k-1|k-1} + Bu_k \tag{3}$$

$$P_{k|k-1} = AP_{k-1|k-1}A^T + Q \tag{4}$$

$$K_k = P_{k|k-1}H^T(HP_{k|k-1}H^T + R)^{-1} \tag{5}$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k(y_k - H\hat{x}_{k|k-1}) \tag{6}$$

where Q and R are the process and measurement noise covariance matrices.

For nonlinear systems, an Extended Kalman Filter (EKF) or Unscented Kalman Filter (UKF) could be used.

D. Digital Twin and Decision-Support Interface

The estimated state feeds the digital twin. The digital twin can then interact with:

- a simulation engine for scenario exploration,
- an AI module for policy learning or recommendation generation,
- a dashboard for human decision-makers.

Task for Daniel: implement the estimator and define the exact data flow from real observations to estimated state.

Comment for Omar: review whether the paper should emphasize estimation accuracy, architectural contribution, or industrial decision-support value.

V. System Architecture

Figure 1 illustrates the conceptual architecture of the proposed system.

Sensors / SCADA → Data Processing → State Estimation → Digital Twin → Simulation / AI Decision Layer → Operational Dashboard

Fig. 1. Conceptual architecture of the proposed framework.

In the broader project, other team members may develop the data backend, simulation engine, AI reasoning/optimization module, and frontend application. The present paper focuses specifically on the **state estimation and digital twin grounding layer**, which is the responsibility of the student author.

VI. Case Study Design

A case study should be defined using either:

1) a real mining process with available data, or
2) a synthetic but realistic industrial scenario inspired by mining operations.

The study should compare at least two conditions:

- decisions or simulations based on raw observations,
- decisions or simulations based on estimated states.

Potential evaluation metrics include:

- state estimation error,
- robustness to sensor noise,

- stability of simulated outputs,
- downstream decision quality,
- computational performance.

Task for Daniel: define the dataset, baseline methods, metrics, and validation protocol.

VII. Expected Results and Hypotheses

The central hypothesis is that a state-estimation-informed digital twin will outperform raw-data-based approaches in terms of consistency, robustness, and decision relevance.

Expected outcomes include:

- reduced discrepancy between observed and inferred process states,
- improved simulation realism,
- better support for downstream optimization or AI-based recommendation,
- greater interpretability for industrial stakeholders.

Comment for Omar: this section can be reframed into formal research hypotheses if the target venue requires it.

VIII. Discussion

This paper is motivated by a broader systems perspective: industrial AI should not be treated purely as a prediction problem. For cyber-physical systems such as mining operations, trustworthy decision support depends on the correct representation of physical reality.

The contribution of this work lies in showing that state estimation is not merely an implementation detail, but a foundational layer that enables higher-level simulation and AI systems to operate on a reliable internal model of the operation.

A key long-term implication is that this architecture can generalize beyond mining, including energy, logistics, autonomous inspection, and industrial robotics.

IX. Conclusion

This paper presented an initial framework for state estimation in mining-oriented digital twins for decision support. By integrating process modeling, sensor data, and estimation techniques, the proposed approach aims to provide a grounded operational state for simulation and AI-based recommendation systems.

This draft is intentionally prepared as a collaborative research starting point. The next steps are to formalize the case study, implement the estimator, validate the system experimentally, and sharpen the scientific contribution for publication.

Collaboration Notes

For Daniel (student):

- Complete the case study definition.
- Implement the model and estimator.
- Add equations, figures, and experiments.
- Expand the references and related work.

For Omar Salinas (professor):

- Comment on scientific novelty and publication angle.
- Improve research framing and literature positioning.
- Suggest target journal/conference and methodological refinements.

For Jorge I. Vásquez (co-author):

- Align the paper with the larger decision-support architecture.
- Contribute industrial/system-level framing.
- Support integration with simulation and AI modules.

Acknowledgment

This draft was prepared as an initial collaborative document to support student-led research in cyber-physical decision-support systems for industrial and mining operations.

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